

Trade, Oligopsony Power, and Wage Inequality*

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Abstract

We show that trade liberalization in oligopsonistic environments can generate heterogeneous markdown responses across different skill groups, thereby exacerbating within-firm wage dispersion. We empirically study this monopsony-driven inequality channel using a quasi-experiment provided by Taiwan's Economic Cooperation Framework Agreement (ECFA) with China, which liberalized trade for a selective set of product categories. Focusing on Taiwanese machinery manufacturers, we find that the ECFA increased overall wages by 7%. However, it raised wage markdowns by 6.9% over low-skilled workers, while markdowns over high-skilled workers were unaffected. The resulting heterogeneous markdown responses aligns with an increase of 7.5%–10.4% in within-firm wage dispersion.

Keywords: *Trade Liberalization, Labor Market Power, Wage Markdown, Within-Firm Wage Dispersion*

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I Introduction

Understanding how trade disciplines firm market power and facilitates efficiency gains is a central question in international economics. A large literature highlights that trade liberalization typically reallocates resources from less to more productive firms and exposes productive firms to greater global competition, mitigating inefficiencies from firms' product market power (Melitz, 2003; Melitz and Ottaviano, 2008; Holmes, Hsu, and Lee, 2014; Edmond, Midrigan, and Xu, 2015; Arkolakis, Costinot, Donaldson, and Rodríguez-Clare, 2019). This reallocation generates the so-called “pro-competitive gains from trade.”

The same reallocation mechanism, however, can have adverse consequences in labor markets. When firms exert monopsony power, concentrating production in a few top firms strengthens their ability to suppress factor payments, diminishing efficiency gains. Moreover, heterogeneous effects on markdowns across factors create important *distributional* consequences, raising inequality in factor returns within top firms. While recent theoretical research has emphasized this *monopsony-driven inequality* channel in closed economies (Deb, Eeckhout, Patel, and Warren, 2024), its implications in an open-economy context remain unexplored, and empirical evidence is scarce. This paper addresses this gap by studying how trade liberalization amplifies such monopsony-driven inequality both theoretically and empirically.

To guide our empirical analysis, we start by developing an open-economy model with oligopoly in product markets and oligopsony in labor markets, building on recent trade-labor frameworks (Edmond, Midrigan, and Xu, 2015; Berger, Herkenhoff, and Mongey, 2022; Gutiérrez, 2022; Deb, Eeckhout, Patel, and Warren, 2024). The comparative statics exercise predicts that trade liberalization reallocates resources toward productive exporters, raising their wage

markdowns. Because high-skilled workers are clustered in these firms prior to liberalization, while low-skilled workers are more evenly distributed, the post-liberalization reallocation disproportionately affects the low-skilled. This uneven reallocation leads to a greater rise in markdowns for low-skilled labor, amplifying within-firm wage inequality among exporters. This identifies a novel channel through which trade reshapes within-firm wage inequality via heterogeneous responses of markdowns across skill groups.

We test these predictions using novel firm-worker-customs matched data on Taiwanese machinery and equipment manufacturers (2006–2017). Following the *production approach* (Dobbelaere and Mairesse, 2013; Yeh, Macaluso, and Hershbein, 2022), we estimate wage markdowns for high- and low-skilled labor without labor market conduct assumptions. We combine the structural estimates with the quasi-experimental variation induced by the Economic Cooperation Framework Agreement (ECFA) between Taiwan and China, announced in 2010. This tariff liberalization provides a uncommonly clean quasi-experimental setting, as its swift implementation and exogenous product scope precluded firm anticipation and strategic selection. Our strategy compares “treated” exporters of ECFA products to China with “control” exporters of the same products to other destinations, applying a doubly robust difference-in-differences estimator (Sant’Anna and Zhao, 2020; Callaway and Sant’Anna, 2021; Caetano and Callaway, 2024) to account for pre-existing differences. This design allows us to identify the causal impact of trade liberalization on markdowns, payrolls, and within-firm inequality.

Our results align with the model’s predictions. Exporters exposed to the ECFA increase wage markdowns for low-skilled labor by 6.9% relative to the control group, while markdowns for high-skilled labor remain unchanged. Average employee pay rises by 7%, reflecting higher

export sales, but workers remain underpaid relative to their marginal product. Crucially, the ECFA also widened within-firm wage dispersion: the coefficient of variation in wages among treated exporters rose by up to 10.4% compared to controls, qualitatively consistent with the heterogeneous markdown response predicted by our model.

Our paper connects and contributes to several strands of literature. First, it relates to works studying the welfare effects of trade liberalization in imperfectly competitive markets ([Melitz and Ottaviano \(2008\)](#), [Edmond, Midrigan, and Xu \(2015\)](#), [De Loecker, Goldberg, Khandelwal, and Pavcnik \(2016\)](#), [Arkolakis, Costinot, Donaldson, and Rodríguez-Clare \(2019\)](#), [Lu, Sugita, Zhu, et al. \(2019\)](#), [Felix \(2021\)](#), [MacKenzie \(2021\)](#), [Gutiérrez \(2022\)](#), [Pham \(2023\)](#), [Xie, Xu, and Yu \(2024\)](#)). Most closely related, [Felix \(2021\)](#) develops a parametric empirical model of labor supply as our conceptual framework, and shows that trade liberalization in Brazil increased market concentration, raising wage markdowns enough to offset modest wage gains from reallocation. [Gutiérrez \(2022\)](#) builds a general equilibrium model with market power in both product and labor markets to study the pro-competitive gains from trade, carefully identifying the boundaries of local markets. Complementary to these seminal works, our empirical approach follows [Yeh et al. \(2022\)](#) and [Rubens \(2023\)](#) in estimating markdowns empirically without imposing strong parametric assumptions about the structure of labor markets, while also incorporating skill-biased technologies (embodied in imported capital) as emphasized by [Burstein, Cravino, and Vogel \(2013\)](#), [Rubens, Wu, and Xu \(2024\)](#), and [Fan and Li \(2025\)](#). Most importantly, the unique features of our dataset allow us to recover markdowns separately for different skill groups and to examine whether their disproportionate responses to trade liberalization amplify within-firm wage inequality.

To the best of our knowledge, this paper is the first to highlight the mechanism of monopsony-driven inequality in an open economy. In doing so, it also connects to the broader literature on globalization and inequality ([Helpman, Itskhoki, and Redding \(2010\)](#), [Sampson \(2014\)](#), [Helpman, Itskhoki, Muendler, and Redding \(2016\)](#), [Grossman, Helpman, and Kircher \(2017\)](#), [Burstein and Vogel \(2017\)](#), [Trottner \(2019\)](#), [Bernard, Fort, Smeets, and Warzynski \(2020\)](#), [Arni, Egger, Erhardt, Gubler, and Sauré \(2024\)](#)). Several works, such as [Helpman et al. \(2010\)](#) and [Helpman et al. \(2016\)](#), emphasize the selection effects of trade. When combined with firms' wage-setting power, selection generates wider wage inequality between firms. [Sampson \(2014\)](#) and [Grossman, Helpman, and Kircher \(2017\)](#) further incorporate worker heterogeneity and examine how trade openness influences the matching and sorting of workers to heterogeneous firms, thereby affecting the distributional consequences both between and within firms. Our work distinguishes itself by focusing on within-firm wage inequality arising from heterogeneous markdown responses across skill groups.

Lastly, our paper speaks to the literature that studies unpacking the secular trend in firm market power and the emergence of superstar firms ([Autor, Dorn, Katz, Patterson, and Van Reenen \(2020\)](#), [De Loecker, Eeckhout, and Unger \(2020\)](#), [Berger, Herkenhoff, and Mongey \(2022\)](#), [Deb, Eeckhout, Patel, and Warren \(2024\)](#)). In our paper, we highlight the contribution of globalization to such a secular trend by leveraging novel firm-worker-customs matched data together with a quasi-experiment of the ECFA.

The remainder of the paper is structured as follows. Section [II](#) presents a conceptual framework that illustrates the effect of trade on wage markdowns and within-firm wage dispersion. Section [III](#) describes the data and empirical setting, estimates the firm-level wage markdowns,

and investigates the causal impact of trade liberalization on wage markdowns and within-firm wage dispersion. Section IV concludes.

II Conceptual Framework

In this section, we present a simple model to illustrate the core mechanism of the paper, providing theoretical guidance for our empirical analysis. In the model, trade liberalization facilitates the reallocation of resources toward productive exporting firms, thereby widening between-firm wage inequality. In oligopsonistic labor markets, this reallocation increases wage markdowns for exporters while reducing them for non-exporters. Moreover, due to skill-biased technology, the reallocation effect is disproportionately stronger for low-skilled workers. Compared to their high-skilled counterparts, low-skilled workers experience a larger increase in wage markdowns among exporters and a larger decrease among non-exporters. This in turn raises within-firm wage inequality among exporters while alleviating it among non-exporters.

II.I An Illustrative Parametric Example

We consider an open economy with two identical countries. In each country, there exists a representative household and heterogeneous firms. The household supplies low- and high-skilled labor in oligopsonistic labor markets and consumes goods produced by firms in oligopolistic product markets. We adopt the following market structure as in [Atkeson and Burstein \(2008\)](#), [Edmond, Midrigan, and Xu \(2015\)](#), [Berger, Herkenhoff, and Mongey \(2022\)](#), [Gutiérrez \(2022\)](#), and [Deb, Eeckhout, Patel, and Warren \(2024\)](#). There is a continuum of identical markets with mass I . Each market $i \in I$ accommodates a finite number of firms, indexed by $n \in \{1, 2, \dots, N\}$.

Each firm hires both skills and produces a differentiated product, taking into account the residual labor supply curves and product demand curve. In what follows, we focus on the Home country, and all the foreign variables will be denoted with a superscript star (*).

Preferences. The Home household maximizes the following static utility:

$$\max_{C_{ni}, L_{ni}, H_{ni}} C - \frac{1}{\bar{\zeta}_L} \frac{L^{1+\zeta_L}}{1+\zeta_L} - \frac{1}{\bar{\zeta}_H} \frac{H^{1+\zeta_H}}{1+\zeta_H},$$

$$s.t. PC = W_L L + W_H H + \Pi,$$

with $\bar{\zeta}_S, \zeta_S > 0, S \in \{H, L\}$. P, W_L, W_H denote the price index of consumption good, and wage indices of low- and high-skill labor. C, H, L are the Constant Elasticity of Substitution (CES) aggregator of consumption, labor supply across firms within markets and across markets. That is,

$$C = \left(\int_{i \in I} I^{-\frac{1}{\theta}} C_i^{\frac{\theta-1}{\theta}} di \right)^{\frac{\theta}{\theta-1}}, C_i = \left(\sum_{n \in N} C_{ni}^{\frac{\gamma-1}{\gamma}} + \sum_{n \in N^*} C_{ni}^{*\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma}{\gamma-1}},$$

$$S = \left(\int_{i \in I} I^{\frac{1}{\theta_S}} S_i^{\frac{\theta_S-1}{\theta_S}} di \right)^{\frac{\theta_S}{\theta_S-1}}, S_i = \left(\sum_{n \in N} S_{ni}^{\frac{\gamma_S-1}{\gamma_S}} \right)^{\frac{\gamma_S}{\gamma_S-1}}.$$

Due to international trade, Home household can also consume the products produced by Foreign firms, C_{ni}^* ¹. Following [Deb, Eeckhout, Patel, and Warren \(2024\)](#), we assume that goods and labor within a market are more substitutable compared to goods and labor across markets, with elasticities $\gamma > \theta > 0$ and $\gamma_S > \theta_S > 0$. We choose the domestic final consumption good C as the numeraire of the economy.

Household utility maximization leads to the following inverse product demand curve and

¹Note that due to fixed cost of exporting, not all the Foreign firms will enter Home market. Hence, $N \neq N^*$. Since we have two identical countries, C_{ni}^* is also the consumption of Foreign household on Home product.

labor supply curves:

$$C_{ni} = \frac{1}{I} P_{ni}^{-\gamma} P_j^{\gamma-\theta} P^\theta C, C_{ni}^* = \frac{1}{I} P_{ni}^{*-\gamma} P_j^{\gamma-\theta} P^\theta C, \quad (1)$$

$$S_{ni} = \frac{1}{I} W_{S,ni}^{\gamma_s} W_{S,i}^{\theta_s-\gamma_s} W_S^{-\theta_s} S, \quad (2)$$

where market price and wage indices are given by

$$P_i = \left(\sum_{n=1}^N P_{ni}^{1-\gamma} \right)^{\frac{1}{1-\gamma}}, P = \left(\int_{i \in I} \frac{1}{I} P_i^{1-\theta} di \right)^{\frac{1}{1-\theta}},$$

$$W_{S,i} = \left(\sum_{n=1}^N W_{s,ni}^{1+\gamma_s} \right)^{\frac{1}{1+\gamma_s}}, W_S = \left(\int_{i \in I} \frac{1}{I} W_{S,i}^{1+\theta_s} \right)^{\frac{1}{1+\theta_s}}.$$

Production. Each firm n in market i produces a differentiated product using the following technology,

$$y_{ni} = \left((z_{L,ni} L_{ni})^{\frac{\sigma-1}{\sigma}} + (z_{H,ni} H_{ni})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}},$$

where $\sigma > 1$ is elasticity of substitution between low- and high-skill labor. $z_{S,ni}$ captures the skill-specific productivity. It's straightforward to incorporate “skill-biased productivity” in our model as in [Burstein and Vogel \(2017\)](#). To see this, we could specify the skill-specific productivity to be $z_S = z \Phi_S$, where z is the Hicks-neutral productivity, and the relative productivity of skills $\frac{\Phi_H}{\Phi_L}$ is increasing in z .

Since the economy consists of a continuum of markets, each firm is small relative to the whole economy. However, within each market, there are finite number of firms. Hence, firms engage in oligopsonistic and oligopolistic competition with each factor and product market, and maximize profit taking into the product demand curve and labor supply curves given in Equations (1) and (2). Following [Melitz \(2003\)](#), we assume that there exist both (iceberg)

trade cost $\tau > 1$ and fixed cost $F_x > 0$ for a firm to serve the Foreign household. In order to deliver 1 unit of product, τ units need to be shipped. Part of the trade cost reflects physical shipping cost, while the rest reflects man-made trade policies such as tariff. We assume that fixed cost F_x is in the unit of final consumption good.

Formally, firm's profit maximization problem is given by

$$\begin{aligned} \Pi_{ni} = \max_{H_{ni}, L_{ni}, q_{ni}, q_{ni}^*, 1_{x,ni} \in \{0,1\}} & P_{ni} q_{ni} + 1_{x,ni} P_{ni}^* q_{ni}^* - W_{H,ni} H_{ni} - W_{L,ni} L_{ni} - 1_{x,ni} F_x, \\ \text{s.t. (1), (2), } & q_{ni} + \tau 1_{x,ni} q_{ni}^* = \left((z_{L,ni} L_{ni})^{\frac{\sigma-1}{\sigma}} + (z_{H,ni} H_{ni})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \end{aligned}$$

where $1_{x,ni} \in \{0, 1\}$ indicates firm's decision of whether to export and serve the Foreign country, and q_{ni}^* is the corresponding quantity received by the Foreign household. Optimal decision leads to the following optimal demand of skill S labor,

$$\frac{P_{ni} y_{ni}^{\frac{1}{\sigma}} z_{S,ni}^{\frac{\sigma-1}{\sigma}} S_{ni}^{-\frac{1}{\sigma}}}{\mu_{ni}} = W_{S,ni} \delta_{S,ni}, \quad (3)$$

where marginal revenue product of labor equates marginal cost of labor. Due to market power, firms internalize the effect of their own quantity choices on their prices and wages. This leads to product markup and wage markdown given by μ_{ni} and $\delta_{S,ni}$. They are closely connected to firm's within-market "market share", and are given by

$$\mu_{ni} = \frac{1}{1 - [\frac{1}{\theta} x_{ni} + \frac{1}{\gamma} (1 - x_{ni})]}, \quad (4)$$

$$\delta_{S,ni} = 1 + \frac{1}{\theta_S} e_{S,ni} + \frac{1}{\gamma_S} (1 - e_{S,ni}). \quad (5)$$

x_{ni} denotes firm n 's domestic sales share in market i , $x_{ni} = \frac{p_{ni}q_{ni}}{P_i C_i}$, and $e_{S,ni}$ denote firm n 's wage bill share of skill S in market i , $e_{S,ni} = \frac{W_{S,ni}S_{ni}}{W_i S_i}$. Given that within-market elasticity is larger, firms with larger product market share and factor market share have larger markups and markdowns. For example, consider a firm with market shares $x_{ni} \approx 1$, $e_{S,ni} \approx 1$. Such firm has the largest markup $\frac{\theta}{\theta-1}$ and markdown $1 + \frac{1}{\theta_s}$. Since the firm controls the whole market, its markup and markdown are only related to across-market substitution.

If firm chooses to serve Foreign country, optimal quantity q_{ni}^* must be such that

$$\frac{P_{ni}}{\mu_{ni}} = \frac{1}{\tau} \frac{P_{ni}^*}{\mu_{ni}^*}. \quad (6)$$

where μ_{ni}^* denotes firm's markup in the Foreign market². The intuition of Equation (6) is that firm allocates quantities in Home and Foreign markets such that marginal revenue product of labor are equalized³. Firm will choose to export if under this optimal quantity q_{ni}^* , the net profit of serving Foreign country is larger than the fixed cost of exporting F_X .

Finally, implied by Equation (3), within-firm wage inequality satisfies:

$$\ln\left(\frac{W_{H,ni}}{W_{L,ni}}\right) = \ln\left(\frac{\delta_{L,ni}}{\delta_{H,ni}}\right) + \frac{\sigma-1}{\sigma} \ln\left(\frac{z_{H,ni}}{z_{L,ni}}\right) - \frac{1}{\sigma} \ln\left(\frac{H_{ni}}{L_{ni}}\right). \quad (7)$$

It depends on the relative productivity of skills, as well as the wage markdowns in our oligopsonistic setup. If, for example, low-skilled labor exhibits larger wage markdown relative to high-skilled labor, this increases the within-firm wage inequality.

²That is, $\mu_{ni}^* = \frac{1}{1 - [\frac{1}{\theta} x_{ni}^* + \frac{1}{\theta_s}(1 - x_{ni}^*)]}$ and $x_{ni}^* = \frac{P_{ni}^* q_{ni}^*}{P_i C_i}$.

³Notice that without endogenous markup, $P_{ni}^* = \tau P_{ni}$ as in a monopolistic competition environment such as Melitz (2003).

Equilibrium. The equilibrium of the economy can be defined in terms of allocations $\{q_{ni}, q_{ni}^*, L_{ni}, H_{ni}\}_{n,i}$ and prices $\{P_{ni}, P_{ni}^*, W_{H,ni}, W_{L,ni}\}_{n,i}$ such that they satisfy household's and firms' first-order conditions, and all the markets clear.

II.II The Effect of Trade Liberalization

When markets undergo trade liberalization with lower tariffs (smaller trade cost τ), focusing on the Home country, two major effects arise. First, within each Home market, the entry of foreign products intensifies product market competition. Second, Home firms gain the potential to access foreign households and earn exporting profits. Due to the fixed cost of exporting, these effects lead to selection of firms (Melitz (2003)) and resource reallocation. Productive firms are able to overcome the fixed cost and expand their production scales and profits into foreign markets, whereas unproductive firms serve only domestic markets and experience declining profit margins due to increased foreign competition.

Resource reallocation has important implications for wage markdowns and within-firm wage inequality, as implied by Equation (7). Following trade liberalization, exporters expand their factor market shares and exert larger wage markdowns, while wage markdowns of non-exporters decline. The magnitude of these responses, however, differs across skill levels. Due to skill-biased productivity, high-skilled workers are already concentrated among productive firms, whereas low-skilled workers are more evenly distributed. Consequently, trade liberalization induces a stronger reallocation in the low-skilled labor market. Compared to their high-skilled counterparts, low-skilled workers experience a larger increase in wage markdowns among exporters and a larger decrease among non-exporters. This ultimately raises within-firm

wage inequality for exporters while reducing it for non-exporters following trade liberalization. We view this as a new mechanism that trade liberalization can lead to higher within-firm wage inequality for exporters, compared to non-exporters⁴.

Numerical example. To illustrate the mechanism qualitatively, we construct the following numerical example. We consider a continuum of markets with mass one. In each market, there are two firms ($N = 2$) competing with each other in an oligopolistic fashion in both product and labor markets. Firm’s production technology exhibits skill-biasedness. Without loss of generality, we assume that firm 2 is more productive, and demands disproportionately more high-skilled labor. We select productivity such that in the high-tariff trade equilibrium, firm 2 employs 95% of high-skilled labor within a market, while only 65% of low-skilled labor. We posit that products and labor are more substitutable within the sector.⁵ We specify fixed cost F_x such that only firm 2 is willing to serve Foreign market. We first solve the high-tariff trade equilibrium. We then decrease the tariff (embedded in τ) to reflect trade liberalization, and re-solve the low-tariff trade equilibrium.

Panel A of Figure 1 illustrates the effect of trade liberalization on wages. For illustrative purpose, we normalize Firm 1’s low-skilled wage in the high-tariff equilibrium to 1. We observe that under skill-biased productivity, firm 2 offer disproportionately higher wages to high-skilled labor. This becomes increasingly so after trade liberalization as firm 2 further expands the pro-

⁴Under skill-biased productivity, resource reallocation also increase the relative aggregate demand of high-skilled labor. This increases the within-firm wage inequality across all the firms.

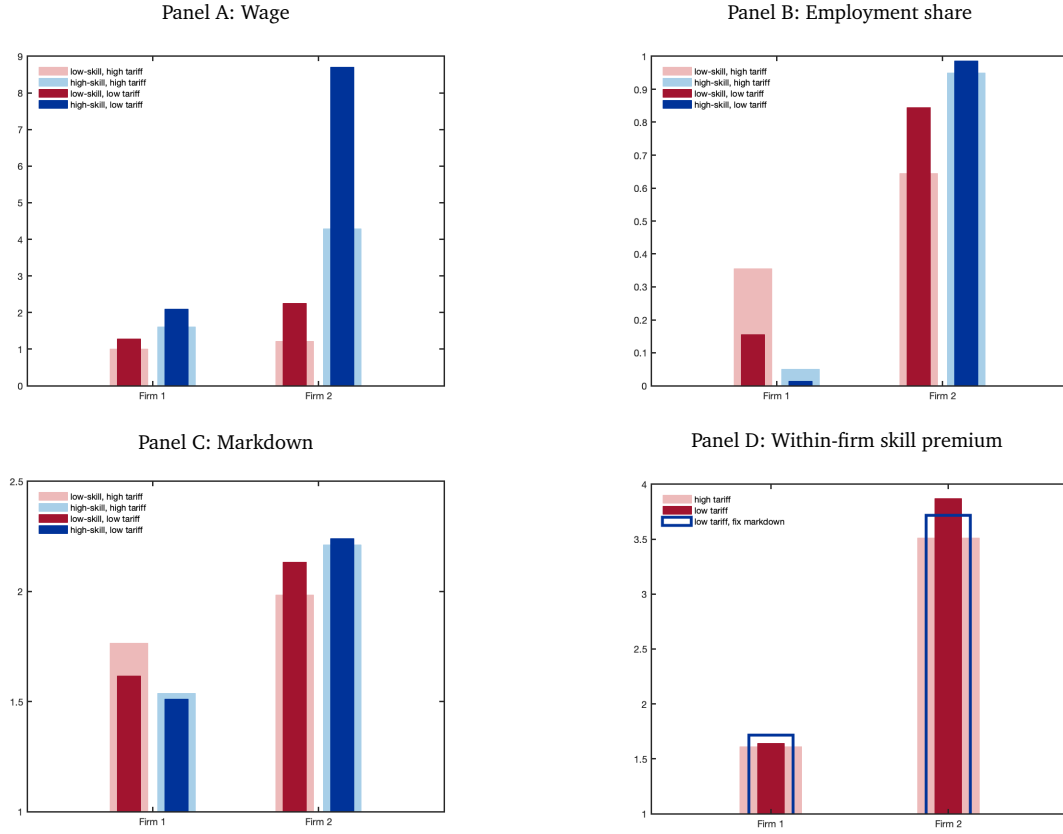
⁵Specifically, we set $\gamma = 6$, a commonly used value for the within-market elasticity of substitution in the international trade literature. Following Edmond et al. (2015), we set $\theta = 1.2$. We take $\theta_s = 0.8$ from Felix (2021), and choose $\gamma_s = 2$ to ensure that, in the high tariff equilibrium, the markdown applied to high-skilled workers by firm 1 is 1.5.

duction scale⁶. Panel *B* shows that there exist resource reallocation towards firm 2 after trade liberalization. However, the effect in the low-skilled labor market is much stronger. The reallocation of factor market share leads to the changes in wage markdowns. From Panel *C*, we observe that markdowns of both skills increase for firm 2, and decrease for firm 1. At the same time, the changes in low-skilled markdown are disproportionately larger. Finally, Panel *D* illustrates the changes in within-firm wage inequality. At first glance, within-firm wage inequality increases in both firms due to rising aggregate demand of high-skilled labor in the economy; however, the increase is stronger within firm 2. To isolate the role of wage markdowns, we examine the hypothetical changes in within-firm wage inequality when wage markdowns are held fixed at their high-tariff equilibrium levels. Under this scenario, we observe that wage inequality within firm 2 declines substantially, while it increases within firm 1. This highlights that our proposed mechanism, where heterogeneous wage markdown responses influence within-firm wage inequality, could be important for quantitative exploration. In Appendix [A](#), we further vary the degree of product market power, as it influences the magnitude of resource reallocation following trade liberalization. We find similar qualitative responses of wage markdowns.⁷

⁶Firm 1's wages also increase relative to the price of the final consumption good, as trade liberalization increases the real wage of the economy.

⁷In Appendix [A](#), we also illustrate the responses of product markups to trade liberalization. We find that domestic markups decline for both firms due to increased foreign import competition, consistent with the findings in [Edmond et al. \(2015\)](#). In contrast, Firm 2's markup in the foreign market rises following trade liberalization, as shown in [Loecker and Warzynski \(2012\)](#).

Figure 1: A Qualitative Example: The Effect of Trade Liberalization



Note: The figure displays the effect of trade liberalization on wage, employment share, wage markdown and within-firm skill premium. Firms are ranked according to their productivity level. The baseline equilibrium involve high tariff. The trade liberalization equilibrium refers to the one with low tariff.

III Empirical Application

III.I Institutional Backgrounds

Our empirical application focuses on the impact if the Economic Cooperation Framework Agreement (ECFA) on the labor market power of Taiwanese machinery and equipment manufacturing firms from 2006 to 2017. Below, we provide an overview of the ECFA and the Taiwanese machinery and equipment sector.

Economic Cooperation Framework Agreement

Following accession to the World Trade Organization (WTO) in 2001, China emerged as an important market for global manufacturers.⁸ The increasing prominence of Chinese final goods manufacturers prompted the development of regional trade agreements (RTAs) to strengthen trade relations with China. Key agreements include the ASEAN-China Free Trade Agreement (CAFTA), implemented in January 2010, and the Regional Comprehensive Economic Partnership (RCEP), which began formal negotiations in 2011.

The establishment of CAFTA posed significant competitive challenges for Taiwanese manufacturers in the Chinese market, as Taiwan lacked preferential tariff agreements with China. To address this, the Taiwanese government initiated negotiations for a bilateral trade agreement with China in 2010. This effort culminated in the signing of the Economic Cooperation Framework Agreement (ECFA) in September 2010, following rapid negotiations that began earlier that year.

To mitigate potential political and economic concerns associated with rapid trade liberalization, the ECFA included an initial phase of tariff reductions targeting a limited set of products (referred to as Early Harvest Products) prior to broader liberalization. However, the progress of the agreement was interrupted in 2014 due to Taiwan's Sunflower Movement, resulting in the continued liberalization of only early harvest products.⁹

The Early Harvest Product list consisted of 539 Taiwanese exports to China and 267 Chinese exports to Taiwan. Notably, the selection of Taiwanese exports receiving preferential tariff

⁸This subsection draws from [Hong and Yang \(2011\)](#) and [Chou \(2009\)](#).

⁹See Michael Turton, "The economic legacy of Taiwan's Sunflower movement," *Taipei Times*, March 28, 2022; James X. Morris, "Brian Hioe: The Sunflower Movement, 4 Years Later," *The Diplomat*, July 18, 2018.

reductions mirrored the liberalized product list under CAFTA, thereby restoring the competitiveness of Taiwanese goods in the Chinese market vis-à-vis ASEAN exporters. The agreement significantly boosted Taiwan's aggregate exports to China, which increased by 35% within the first three years of implementation. In contrast, Taiwan's import volume from China remained relatively unchanged during the same period.

Overall, the ECFA event provides a unique quasi-experimental setting for economic analysis. First, the agreement's rapid negotiation and implementation rendered its announcement unanticipated by Taiwanese firms. Second, the selective liberalization of specific products allows for a clear division of firms into treated and control groups within narrowly defined industries. Finally, the alignment of the product selection process with the CAFTA liberalized product list minimized the potential for strategic manipulation by either Taiwan or China.

Taiwanese Machinery and Equipment Industry

Among the 539 Early Harvest Products from Taiwan to China, the largest categories were chemical materials (88 items), machinery and equipment (107 items), and textiles (137 items). We aim to focus on machinery and equipment manufacturers.

The choice of this industry is motivated by its significant benefit from the ECFA. Even prior to the agreement, China was Taiwan's largest export market for machine tools, highlighting the critical role of tariff reductions for this sector. Following the implementation of the ECFA, the machinery and equipment industry experienced a dramatic 603.5% increase in total export sales to China over a decade.¹⁰

¹⁰According to the Ministry of Economic Affairs, Taiwan, machinery and equipment exports to China rose from USD 1.1 billion in 2009 to USD 6.9 billion in 2019.

Additionally, machinery and equipment manufacturers serve a broad range of other manufacturing sectors due to the nature of their products. In contrast, chemical materials and textiles primarily cater to specific downstream industries, such as the paint and apparel/footwear sectors. This unique characteristic enables us to isolate the impact of export tariff liberalization on the industry without concerns about confounding effects from complex global supply chain networks.

III.II Data

We use a comprehensive and unique dataset from the Fiscal Information Agency, Ministry of Finance, Taiwan. This dataset integrates multiple rich sources of information, including (1) business tax filings, (2) individual tax filings, and (3) trade custom records. Covering the universe of profit-seeking firms in Taiwan from 2006 to 2017, along with their international transaction records, this dataset offers an unparalleled opportunity to examine the dynamics of a firm's labor market power in response to trade liberalization.

Business tax returns include balance sheets with unique business identification numbers common across years and filing forms. From these sources, we construct a complete panel of firms with the information on their receipts, material input expenditures, and fixed assets (i.e., plant, property, and equipment; PP&E), which are necessary to estimate the production function. The module records the total payroll allocated to production workers for manufacturing processes. We use this variable to measure unskilled labor input as it can account for the quality of production workers. Additionally, the business tax filings provide 6-digit industry codes, allowing us to classify a firm's operating industry based on the first two digits of their code.

The individual earnings data provide details on compensation received from business entities, including salaries and stock-option bonuses. Using unique business identification numbers, we construct firm-level metrics such as the number of employees and total salary payrolls from the business tax return data. We measure skilled labor by calculating total payrolls for non-production workers (e.g., salespersons and executives), obtained by subtracting payrolls for production workers from total salary payrolls. We also use the data to measure a firm's payroll structure during a given period, including average employee wage rates and wage dispersion.

The trade customs records encompass all international transactions from Taiwan at the HS six-digit code, firm, destination, and year level. This transaction-level panel enables us to identify (1) exporters of ECFA-liberalized products and (2) whether these firms exported such products to China, allowing us to construct a sample of Taiwanese firms directly exposed to the ECFA (i.e., treated firms).

Sample Construction

Our sample construction procedure begins by cleaning business tax returns and retaining only firms that report positive values for net receipts, material expenditures, property, plant, and equipment (PP&E). These criteria ensure the dataset focuses on economically active firms engaged in meaningful operations. Individual tax records are filtered to include only those with annual earnings exceeding the Taiwanese minimum wage. In keeping with the literature ([Lamadon et al. \(2022\)](#)), we define the primary employer for these workers as the business identification number responsible for the highest salary payrolls. We aggregate the trade custom records to the firm-year level with separate values for total export revenue, China export

revenue, ECFA export revenue, and ECFA-China export revenue.

Sales revenue, salary payrolls, and material expenditures are deflated using the industry wholesale price deflator, while PP&E is deflated using the investment deflator. We focus on firm-year observations consistently reported across all dataset modules from 2006 to 2017, resulting in an unbalanced panel of 77,667 firm-year observations. Firms with deflated sales revenue below 5 million NTD and fewer than five employees are excluded. Following [Ruhl and Willis \(2017\)](#), we also remove outlier firms with extreme changes in labor employment and sales revenue. After these filters, the dataset includes 42,128 firm-year observations. For production function estimation, we further refine the sample by trimming the 2nd and 98th percentiles based on revenue-to-variable cost ratios, material shares, and labor-to-material expenditure ratios. This trimming yields an estimation sample of 38,610 firm-year observations.

One of our primary objectives is to assess ECFA's impact on within-firm wage dispersion, which is an empirical assessment of the prediction by our conceptual framework: If the mark-down responses to the ECFA are more substantial for unskilled labor, a within-firm wage dispersion would increase. In this light, we measure wage dispersion by using two metrics derived from employer-employee matched data: (1) the standard deviation of logged payrolls (SDL) and (2) the coefficient of wage variation (CV):

$$V_{jt}^{SDL} = \sqrt{\frac{1}{N_{jt}} \sum_{ijt} (\ln W_{ijt} - \overline{\ln W}_{jt})^2}, \quad (8)$$

$$V_{jt}^{CV} = \frac{\sigma_{jt}^W}{\mu_{jt}^W}, \quad (9)$$

where W_{ijt} refers to payroll for worker i in firm j and year t ; $\overline{\ln W}_{jt}$ is the within-firm average

of logged payrolls; σ_{jt}^W and μ_{jt}^W are the within-firm standard deviation and average of payrolls, respectively.

Table 1 describes the variables of interest in our sample. The sample reveals that the size variables of firms—sales, employees, material expenditures, PP&E, direct labor cost—are highly left-skewed, echoing the stylized facts that have been well demonstrated.

Table 1: Sample Description:
Machinery Industry (Observations = 42,128)

	Median	Mean	Std. Dev
Number of employees	14	32.7853	85.6898
Payroll to Non-production worker*	1.5384	7.3073	25.1940
Payroll to Production worker*	3.0673	7.9832	31.1068
Average worker payroll*	0.3426	0.3678	0.1430
Material expenditures*	19.4356	74.1473	264.7117
Plant, Property, and Equipment*	12.2035	50.7835	256.6563
Sales Revenue*	38.4015	141.3529	489.2294

Notes: The table displays the pooled sample averages of the key variables of interest. Variables marked with an asterisk (*) are expressed in deflated millions of 2016 New Taiwan Dollar.

Sample for Treatment Effect Analysis

The ECFA, in conjunction with the firm-product-level matched panel, allows us to estimate its effects on markdowns and productivity via a difference-in-differences framework. Despite reasonable quasi-experimental variation, a simple comparison of ECFA-target exporters and others post-ECFA after the ECFA is subject to two key identification challenges: (1) firms may

have strategically added ECFA products to their portfolios, and (2) exporters of ECFA items to China differ systematically in size and productivity from those exporting elsewhere.

We thus carry out our treatment analysis focusing on the sample of the firms that ever exported ECFA products. We classify them into treated and control groups based on their history of ECFA export destinations. The treated group consists of firms that consistently exporting ECFA items to China throughout the sample period, directly benefiting from the ECFA tariff reductions. The control group consists of the ECFA exporters that exported ECFA products to countries other than China, while never exporting ECFA items to China. By doing so, we mitigate the biases resulting from (1) the self-selection into export by excluding never-exporting firms and (2) from the strategic product portfolio adjustments post-ECFA. Restricting our analysis to exporters active during 2010 - the year of ECFA implementation - leaves us with a sample of 344 treated and 267 control firms.

First Glance at ECFA

Table 2 presents a summary of trade activities for Taiwanese machinery firms before and after the implementation of the ECFA. The agreement had a notable impact on both the extensive and intensive margins of exporting ECFA items to China. On the extensive margin, the fraction of firms exporting ECFA items to China increased by 4.7 percentage points following the ECFA. On the intensive margin, the average share of export sales of ECFA items to China, relative to total exports to China, rose by 6.68 percentage points. Similarly, the share of ECFA items in total firm-level export sales increased by 1.14 percentage points. These increases were not driven by declining export sales of non-ECFA items but rather by substantial growth in ECFA-China export revenues. Specifically, the average revenue from ECFA items exported to China

grew from 20 million NTD to 34 million NTD, indicating that benefiting firms experienced overall growth in their export performance.

The ECFA also influenced the extensive margins of broader export participation. The fraction of firms exporting ECFA items to any destination rose by 2.86 percentage points, while the fraction of firms exporting to China increased by 1.42 percentage points. Notably, the ECFA encouraged greater participation in the Chinese market, as evidenced by the increase in the share of firms exporting to China. However, the total fraction of exporters remained effectively unchanged, suggesting that the ECFA primarily reallocated existing export activities rather than creating new exporters.

Table 2: Trade before and after ECFA:
Machinery Industry (Observations = 42,128)

	Pre-ECFA	Post-ECFA
Firm-level ECFA-China Export Revenue*	21.3277	34.6161
Firm-level ECFA Item Share of Exports to China	0.5194	0.5860
Firm-level ECFA-China Export Share	0.4307	0.4429
Fraction of Firms Exporting ECFA Items to China	0.2812	0.3286
Fraction of Firms Exporting to China	0.4367	0.4518
Fraction of Firms Exporting ECFA Items	0.5382	0.5674
Fraction of Exporters	0.6613	0.6642

Notes: The table displays the pooled sample averages of the key variables of interest. Revenues are expressed in deflated millions of 2016 New Taiwan Dollar.

* Computed conditional on ECFA-China participation.

The observed improvements in export performance, particularly among firms exporting ECFA items to China, motivate further investigation into how trade liberalization under the ECFA impacted labor market outcomes. To this end, we employ a structural model of firm production to estimate firm-level wage markdowns, which measure the degree of labor market power exerted by firms. By comparing the evolution of wage markdowns for ECFA-China exporters and non-exporters, we aim to assess the labor market power effects of this trade liberalization event. Specifically, we examine whether the increased export revenues associated with the ECFA translated into changes in firms ability to influence wages, thereby shedding light on the broader implications of trade liberalization for labor markets.

Market Concentration Dynamics

Before presenting the formal structural analysis, we examine how the payroll shares of high- and low-skilled labor evolved around the ECFA announcement. We define the local labor market at the citydistrict level and compute payroll market shares for treated and control exporters, separately for low-skilled (production) and high-skilled (non-production) workers.

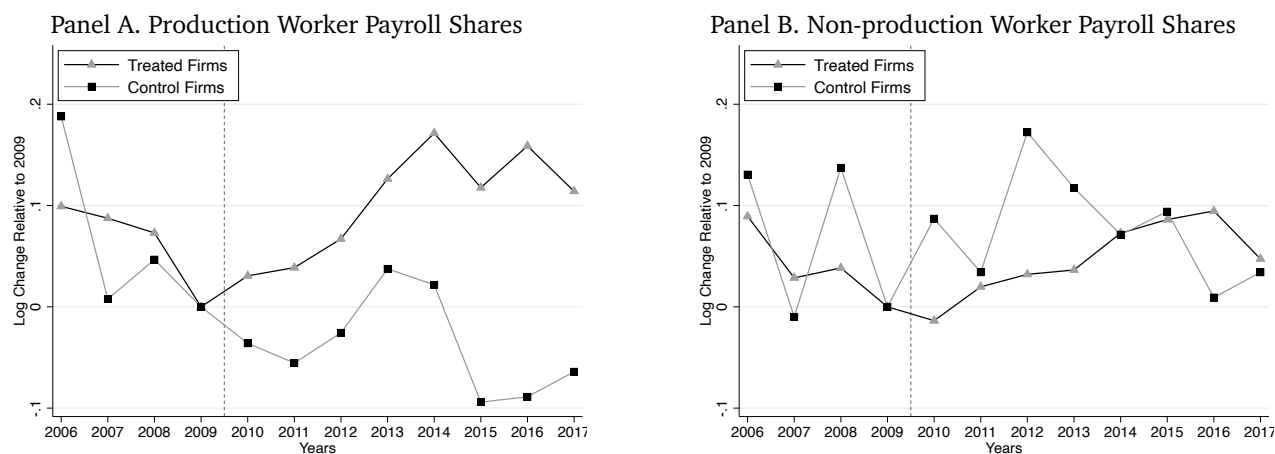


Figure 2: Evolution of Payroll Market Share (Relative to 2009 Level)

Figure 2 shows the evolution of payroll shares, normalized to their 2009 levels. Panel A indicates that both treated and control exporters experienced declines in production worker payroll shares prior to the ECFA. After the ECFA, treated exporters began to recover their shares, suggesting a reallocation of low-skilled labor toward these firms. In contrast, Panel B shows no analogous effect for high-skilled workers, consistent with the comparative-static predictions of our conceptual framework.

Overall, the dynamics of payroll shares suggest that treated exporters may have increased their labor market power over low-skilled workers following the ECFA. However, changes in payroll shares could reflect rising wages rather than the reallocation of low-skilled employment. To determine whether the observed changes reflect labor reallocation and the associated increase in labor market power, we turn to the estimation of wage markdown for low-skilled and high-skilled workers and combine them with the ECFA-induced quasi-experimental variations.

III.III Identification of Markdown and Productivity

We identify wage markdowns at the firm level using *production approaches* in the spirit of Dobbelaere and Mairesse (2013), Yeh et al. (2022), and Rubens (2023). The key advantage of these approaches is that they identify wage markdowns without requiring strong assumptions about labor market conduct. Specifically, the estimation procedure relies on minimal assumptions: firms internalize finitely elastic labor supply and engage in standard optimization processes, namely profit maximization and cost minimization.

The starting point of these approaches is the production function for firm j in period t :

$$Y_{jt} = F_{jt}(L_{jt}, H_{jt}, M_{jt}, K_{jt}), \quad (10)$$

where Y_{jt} denotes output, L_{jt} low-skilled labor, H_{jt} high-skilled labor, M_{jt} raw materials, and K_{jt} capital stock. Firm j is assumed to be a price taker in the raw materials market, while it exerts market power in the output and both labor markets. As outlined in [Yeh et al. \(2022\)](#), profit maximization and cost minimization yield expressions for the firm-specific wage markdowns for high-skilled and low-skilled labor. These markdowns are expressed as the ratio of input expenditures weighted by the respective output elasticities of inputs¹¹:

$$\underbrace{\frac{MPRL_{jt}}{W_{jt}^H}}_{=\delta_{jt}^H} = \frac{\partial \ln Q_{jt} / \partial \ln H_{jt}}{\partial \ln Q_{jt} / \partial \ln M_{jt}} \frac{W_{jt}^M M_{jt}}{W_{jt}^H H_{jt}} \equiv \frac{\theta_{jt}^H}{\theta_{jt}^M} \frac{W_{jt}^M M_{jt}}{W_{jt}^H H_{jt}}, \quad (11)$$

$$\underbrace{\frac{MPRH_{jt}}{W_{jt}^L}}_{=\delta_{jt}^L} = \frac{\partial \ln Q_{jt} / \partial \ln L_{jt}}{\partial \ln Q_{jt} / \partial \ln M_{jt}} \frac{W_{jt}^M M_{jt}}{W_{jt}^L L_{jt}} \equiv \frac{\theta_{jt}^L}{\theta_{jt}^M} \frac{W_{jt}^M M_{jt}}{W_{jt}^L L_{jt}}, \quad (12)$$

where ψ_{jt}^H and ψ_{jt}^L are the elasticities of high-skilled and low-skilled labor supply, respectively. While input expenditures can be obtained directly from accounting data, the output elasticities of input factors must be estimated via the production function.

In our empirical implementation, we consider a translog specification for production tech-

¹¹Detailed derivations are provided in [Appendix B](#).

nology (10):

$$\begin{aligned}
y_{jt} = & \beta_l l_{jt} + \beta_h h_{jt} + \beta_m m_{jt} + \beta_k k_{jt} + \sum_{\{n,o\} \in \{l,h,m,k\}} \sum \beta_{no} n_{jt} o_{jt} \\
& + \left(\beta_l^{SBT} l_{jt} + \beta_h^{SBT} h_{jt} + \beta_m^{SBT} m_{jt} + \beta_k^{SBT} k_{jt} + \sum_{\{n,o\} \in \{l,h,m,k\}} \sum \beta_{no}^{SBT} n_{jt} o_{jt} \right) s_{jt} \\
& + \omega_{jt} + \varepsilon_{jt},
\end{aligned} \tag{13}$$

where y_{jt} , l_{jt} , h_{jt} , m_{jt} , and k_{jt} denote the logs of output, low-skilled labor, high-skilled labor, materials, and capital of Taiwanese manufacturer j in period t . ω_{jt} and ε_{jt} represent Hicks-neutral productivity and output measurement errors, respectively. We also include $s_{jt} \equiv \frac{K_{jt}^{IMP}}{K_{jt}}$, the share of accumulated imported capital stock up to period t in the total capital stock. This interaction term is motivated by empirical evidence showing that imported capital embodies skill-biased technologies and thus disproportionately increases the output elasticity of skilled labor (Burstein et al., 2013; Rubens et al., 2024; Fan and Li, 2025). Accordingly, specification (13) provides a parametric representation of skill-biased technical change (SBTC) by allowing heterogeneous output elasticities of skilled and unskilled labor depending on firms import-capital shares.¹²

We estimate the production function using timing assumptions on input choices and a stronger parametric assumption on the law of motion for the Hicks-neutral productivity term ω_{jt} in (13) (Blundell and Bond, 2000; Akerberg et al., 2015). Specifically, we decompose ω_{jt}

¹²This formulation is conceptually analogous to research on labor-augmenting productivity effects of technology adoption (Foster et al., 2022; Kusaka et al., 2022; Miller et al., 2023).

into a time-invariant component and a time-varying component that follows an AR(1) process:

$$\omega_{jt} = \omega_j^* + \tilde{\omega}_{jt}, \quad (14)$$

$$\tilde{\omega}_{jt} = \rho \tilde{\omega}_{j,t-1} + \xi_{jt}, \quad (15)$$

where ξ_{jt} is an i.i.d. productivity shock. In practice, we construct the associated moment conditions for estimating the parameters using a system-GMM approach (Blundell and Bond, 1998, 2000). A detailed explanation of the implementation is provided in Appendix C.

Discussion on Production Function

Use of Deflated Revenues as Output - It is well established that using revenues as a proxy for output can lead to the underidentification of markups (Bond et al. (2021)). Bond et al. (2021) argue that when revenue is used instead of true output, the estimated production elasticities correspond to revenue elasticities—i.e., the product of price markups and true output elasticities—leading to biased elasticity estimates and uninformative markups. Given that our analysis employs deflated revenues as a measure of output, a similar identification concern may arise regarding our wage markdown estimates.

However, Yeh et al. (2022) demonstrate that the wage markdown equations (11) and (12) are not subject to this bias. Since the bias affects both the revenue elasticities of labor and materials symmetrically, it cancels out in the markdown equation, which is based on their ratio. Even when the bias asymmetrically affects the revenue elasticities, equations (11) and (12) can be interpreted as identifying wage markdowns relative to those for material input markdowns (Treuren (2023)). Specifically, let $\tilde{\theta}_{jt}^V$ denote the revenue elasticity of input V . Treuren

(2023) shows that $\tilde{\theta}_{jt}^L \frac{R_{jt}}{W_{jt}^L L_{jt}}$ and $\tilde{\theta}_{jt}^M \frac{R_{jt}}{W_{jt}^M M_{jt}}$ represent the wage and material input markdowns, respectively. Thus, equations (11) and (12) capture wage markdowns relative to material input markdowns when revenue is used as the output measure. Overall, our estimates provide a lower bound on wage markdowns and the ECFA's impact on them.

Estimation Results

Table 3 reports output elasticities implied by the estimated production function.¹³ Low-skilled labor (θ_{jt}^L) and materials (θ_{jt}^M) account for the bulk of output (0.1977 and 0.6391, respectively), while high-skilled labor (θ_{jt}^H) and capital (θ_{jt}^K) contribute comparatively less.

Addressing endogeneity in input choices and incorporating skill-biased technical changes are crucial in correctly measuring wage markdowns. Compared to ordinary least squares (OLS), our GMM estimates produce smaller material output elasticity, suggesting a potential correction of downward biases in wage markdowns over low-skilled labor based on the OLS-based output elasticities.

¹³Parameter estimates are in Appendix C.

Table 3: Wage Markdowns and Output Elasticity:
Machinery Equipment Industry

	Mean	Std. Dev
Output Elasticity: GMM		
θ_{jt}^L	0.1977	0.0868
θ_{jt}^H	0.0611	0.0304
θ_{jt}^M	0.6391	0.1159
θ_{jt}^K	0.0613	0.0275
Output Elasticity: OLS		
θ_{jt}^L	0.1940	0.0990
θ_{jt}^H	0.0743	0.0236
θ_{jt}^M	0.6612	0.1338
θ_{jt}^K	0.0528	0.0345
Wage Markdowns		
δ_{jt}^L	1.7567	0.7746
δ_{jt}^H	1.0353	0.9402
N	39,133	

Notes: The table displays the summary statistics for estimated wage markdowns and the output elasticities of labor, material, and capital.

The wage markdown estimates highlight the considerable firm's labor market power over low-skilled labor within the Taiwanese machinery industry. The mean wage markdown is estimated at 1.7567, indicating that the mean low-skilled labor in the machinery equipment in-

dustry is paid only 57% of their marginal revenue product. In contrast, the mean high-skilled labor is paid around 97% of their marginal revenue product—indicated by the mean markdown of 1.0353. These different levels of wage suppression align with the global benchmarks. For instance, the median wage markdowns for production and nonproduction workers in the U.S non-electronic machinery sector are 4.530 and 1.359, respectively (Yeh et al. (2022)).

III.IV Liberalization Treatment Effects

Doubly-Robust Differences-in-Difference

Table 4 summarizes the sample averages of logged outcome variables and covariates for treated and control firms from 2006 to 2009. Treated exporters exhibit higher wage markdowns for low-skilled labor (29.9%), higher wage rates (32.2%), and higher TFPR (9.27%) relative to controls. In contrast, both treated and control exporters exhibit similar wage markdowns for high-skilled labor. Lastly, treated exporters show greater within-firm wage dispersion, suggesting a more severe payroll inequality.

Table 4: Treatment Analysis Sample Description:
Averages from 2006 to 2009

	Treated	Control	Difference
Outcome Variables: Performance			
Log Markdowns for Production Worker	0.8102	0.5109	0.2993
Log Markdowns for Non-production Worker	-0.0909	-0.0969	0.0060
Time-varying TFPR	-0.0208	-0.1135	0.0927
Log Average Worker Payroll	-0.7941	-1.1163	0.3222
Outcome Variables: Within-Firm Wage Dispersion			
Log SDL of Wage	-0.7506	-1.0098	0.2592
Log Wage CV	-0.6932	-1.0083	0.3151
Covariates			
Log PP&E (normalized)	1.4002	0.0023	1.3979
Positive Investment Rates	0.1318	0.0971	0.0347
Fixed TFP	0.0001	-0.0166	0.0167
No. Treated firms	344		
No. Control firms	267		

Notes: The table presents the sample average of the outcome variables and control covariates for treated and control exporters. The average is calculated based on the observations prior to the ECFA announcement.

However, significant imbalances in covariates reveal a potential violation of the uncondi-

tional parallel trends assumption critical for difference-in-differences estimation. For instance, treated exporters were three times larger in PP&E, had higher investment rates (by 3.47pp), and exhibited 1.67% greater fixed TFP compared to controls. Despite carefully selecting control firms that also exported ECFA products, these imbalances in key covariates indicate potential threats to the treatment effects identification. The unbalanced covariates imply that treated and control firms may have had differing growth paths independent of ECFA, thus raising concerns about the validity of the unconditional parallel trends assumption.

To address these issues, we estimate the average treatment effect on treated (ATET) using a doubly robust difference-in-differences estimator ([Sant'Anna and Zhao \(2020\)](#)). We construct counterfactual outcomes by weighting control firms based on propensity scores of treatment ([Abadie \(2005\)](#)) and perform regression adjustments for outcomes to mitigate biases from potential misspecification of the propensity score model. We employ the pre-treatment averages of capital stock, investment rates, and fixed TFP as covariates in both propensity score and regression models. The choice of covariates aim to control for differences in size and productivity using key covariates: (1) PP&E to capture firm size, (2) positive investment rates as a proxy for quantity-based productivity ([Olley and Pakes \(1996\)](#)), and (3) fixed TFP to account for innate market capability differences. This approach closely follows [Callaway and Sant'Anna \(2021\)](#) and [Caetano and Callaway \(2024\)](#), with key modifications: (1) ECFA was not introduced staggeredly, so we do not need to account for treatment effect heterogeneity across treatment cohorts, and (2) rather than transforming covariates into time-invariant matrices, suggested by [Caetano and Callaway \(2024\)](#), we use pre-treatment averages for propensity score and outcome adjustments to avoid the curse of dimensionality.

Treatment Effect Estimates

Table 5 presents the estimated effects of the ECFA on wage markdowns for low-skilled labor, wage markdowns for high-skilled labor, and TFPR. The doubly-robust ATET estimates indicate that low-skilled labor (e.g., production workers) of treated exporters receive suboptimal wages. Treated exporters exhibited an increase of 6.9% in wage markdowns for low-skilled labor, reflecting enhanced labor market power over low-skilled labor following the ECFA. In contrast, the ECFA did not affect the labor market over high-skilled labor, echoing the comparative static results in Section II. The estimates further reveal the absence of TFPR premiums for treated exporters.

Table 5: Average Treatment Effect of ECFA (Changes in Log):
Firm Performance

	Wage Markdowns Low Skilled	Wage Markdowns High-skilled	TFPR
ATET	0.069*** (0.021)	0.033 (0.057)	−0.033 (0.025)
No. Treated Firms	344		
No. Control Firms	267		

Notes: The table presents the estimated average treatment effects of the ECFA on wage markdowns for low-skilled labor, wage markdowns for high-skilled labor, and TFP. Block-bootstrapped standard errors are in parentheses. Asterisks indicate levels of statistical significance: * 10%, ** 5%, *** 1%.

We further examine the evolution of markdowns between treated and control exporters after the ECFA, as depicted in Figures 3. Consistent with the ATET estimates in Table 5, ECFA-

treated exporters experienced only increases in markdowns over low-skilled labor. Wage markdowns over low-skilled labor rose substantially over seven years up to 10%. In contrast, wage markdowns over high-skilled labor fluctuate around the ECFA announcement event, suggesting the absence of the labor market effect of the ECFA on high-skilled labor.

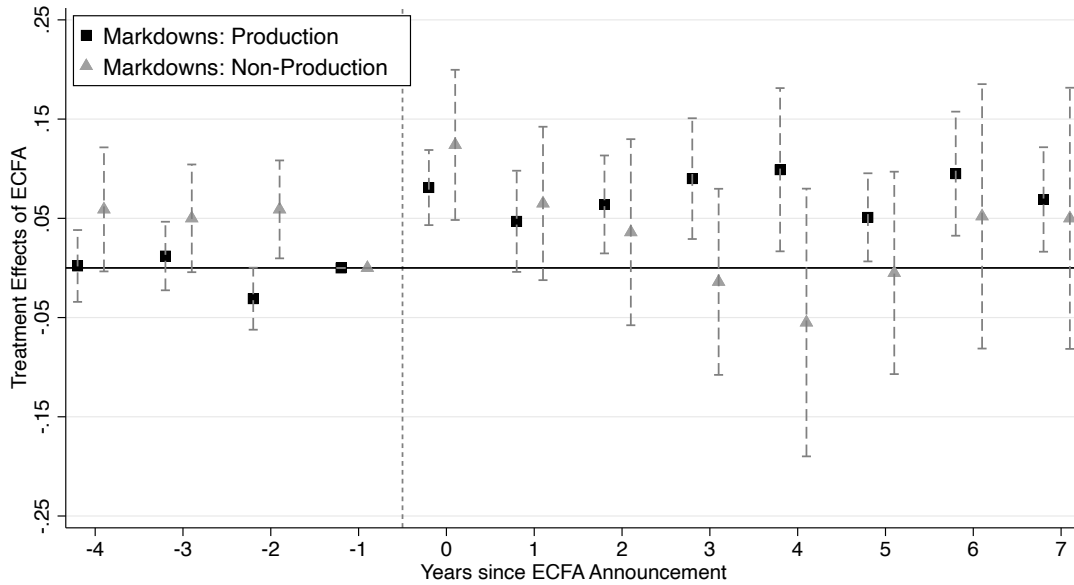


Figure 3: Dynamic Treatment Effects of ECFA: Heterogenous Markdowns

Note. This figure displays the dynamic treatment effects of the ECFA on wage markdowns over low-skilled and high-skilled workers, along with 90% block-bootstrapped confidence intervals. The treatment effects are estimated using the doubly-robust method of [Sant'Anna and Zhao \(2020\)](#), [Callaway and Sant'Anna \(2021\)](#), and [Caetano and Callaway \(2024\)](#).

Lastly, we examine how these labor market power effects influence both the between- and within-firm wage dispersions, as outlined in Section II. In the presence of market power in output and labor markets, trade liberalization can exacerbate wage gaps between firms and but also wage gaps within firms due to heterogeneous responses of wage markdowns. When unskilled labor markdowns increase disproportionately more after trade liberalization, this widens the within-firm wage dispersion. To confirm this prediction, we employ the same doubly-robust difference-in-differences design, applying it to the within-firm wage dispersion

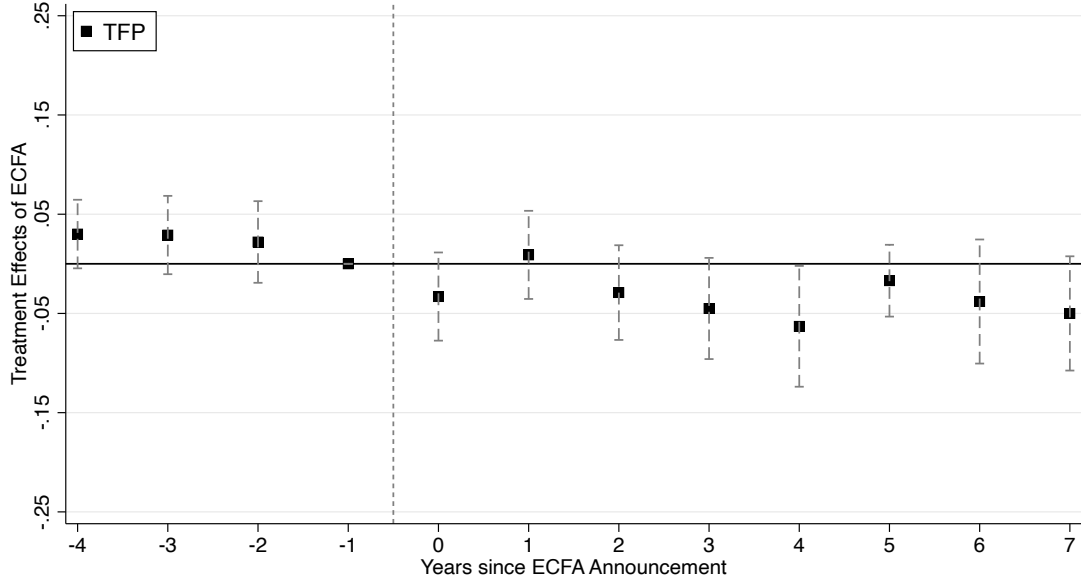


Figure 4: Dynamic Treatment Effects of ECFA: Total Factor Productivity

Note. This figure displays the dynamic treatment effects of the ECFA on TFP, along with 90% block-bootstrapped confidence intervals. The treatment effects are estimated using the doubly-robust method of [Sant’Anna and Zhao \(2020\)](#), [Callaway and Sant’Anna \(2021\)](#), and [Caetano and Callaway \(2024\)](#).

measures calculated in (8) and (9). Increases in these measures indicate a more right-skewed payroll structure, signifying higher within-firm worker inequality.

As shown in Table 6, the doubly robust ATET estimates support the theoretical prediction that trade liberalization increases within-firm wage dispersion. First, the average worker payroll at treated exporters rose by 7%, indicating a widening of wage gaps across firms. In addition, employees of ECFA-treated exporters experienced a 7.5% (10.4%) higher standard deviation (coefficient of variation) of wages compared to those at control firms following the ECFA. Thus, despite receiving wage premiums, employees at treated exporters faced greater wage inequality likely driven by enhanced labor market power. The dynamic treatment effects on wage levels and within-firm wage dispersion, presented in Figures 5 and 6, further corroborate this pattern.

Table 6: Average Treatment Effect of ECFA (Changes in Log):
Payroll Structure

	Wage Rates	SDL of Wage	Wage CV
ATET	0.070***	0.075***	0.104***
	(0.018)	(0.025)	(0.030)
No. Treated Firms	344		
No. Control Firms	267		

Notes: The table presents the estimated average treatment effects of the ECFA on average worker payroll and within-firm wage dispersion measures: the SDL of wage and wage CV. Block-bootstrapped standard errors are in parentheses. Asterisks indicate levels of statistical significance: * 10%, ** 5%, *** 1%.

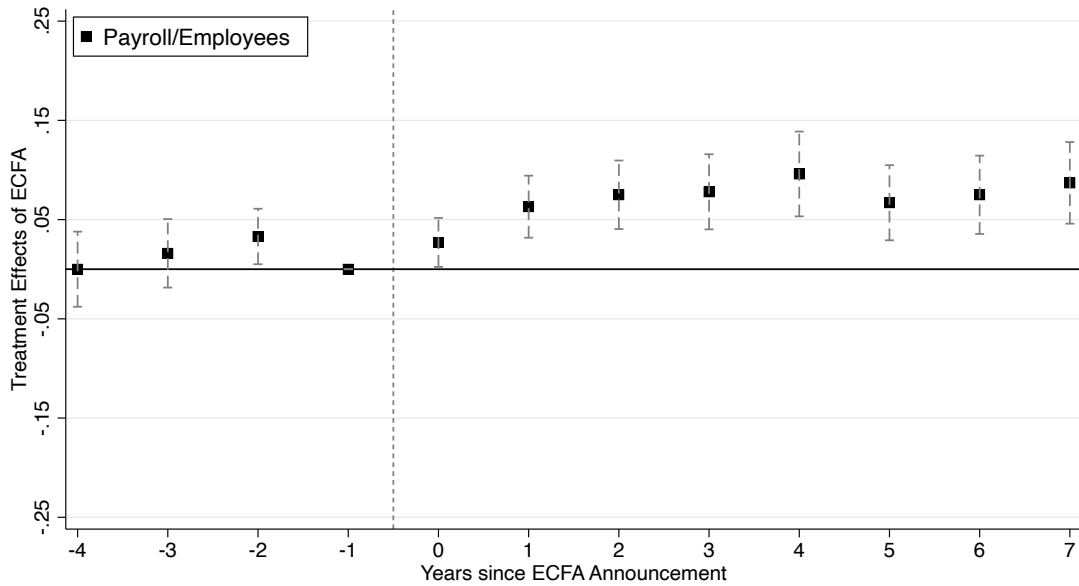


Figure 5: Dynamic Treatment Effects of ECFA: Average Worker Payroll

Note. This figure displays the dynamic treatment effects of the ECFA on the average worker payroll, along with 90% block-bootstrapped confidence intervals. The treatment effects are estimated using the doubly-robust method of [Sant'Anna and Zhao \(2020\)](#), [Callaway and Sant'Anna \(2021\)](#), and [Caetano and Callaway \(2024\)](#).

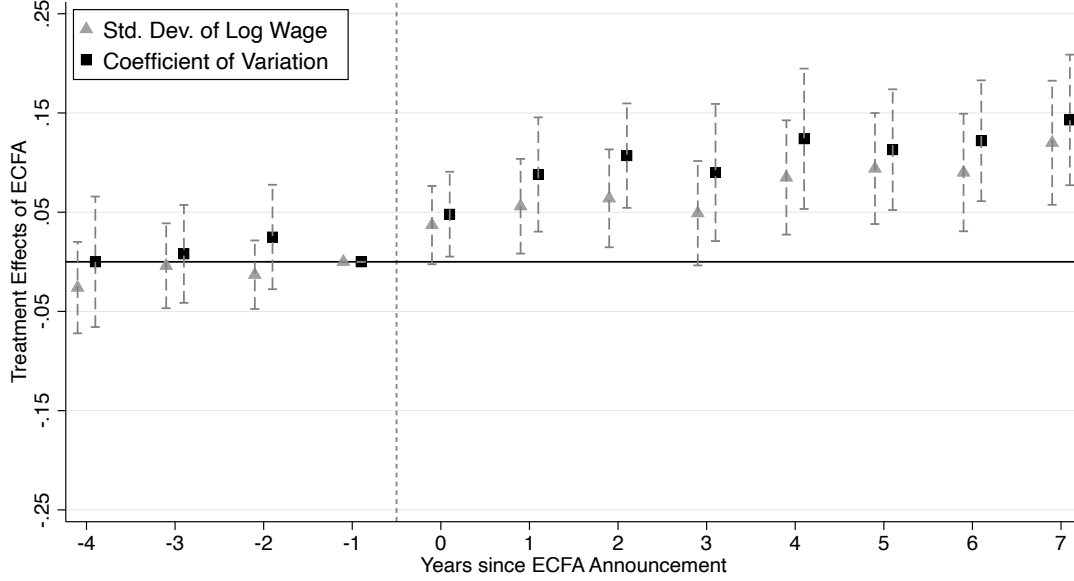


Figure 6: Dynamic Treatment Effects of ECFA: Within-Firm Wage Dispersion

Note. This figure displays the dynamic treatment effects of the ECFA on within-firm wage dispersion, along with 90% bootstrapped confidence intervals. The treatment effects are estimated using the doubly-robust method of Sant’Anna and Zhao (2020), Callaway and Sant’Anna (2021), and Caetano and Callaway (2024).

IV Concluding Remark

In this paper, we analyzed the impact of trade liberalization on labor markets when firms can exercise market power. Resources reallocation towards more productive firms after trade liberalization can increase wage markdowns, and heterogeneous responses of markdowns across different types of labor can exacerbate the within-firm wage dispersion. Our findings revealed that the liberalization of export tariffs on transactions with China led to a substantial 6.9% increase in wage markdowns for low-skilled labor within Taiwans machinery and equipment industry. In contrast, the ECFA did not affect wage markdowns for high-skilled labor. We also demonstrated that the resulting heterogeneous exertion of labor market power of firms following the liberalization contributed to greater within-firm wage dispersion, indicating that workers experienced growing payroll inequality.

Our empirical results offer significant implications that extend beyond the Taiwan-China Free Trade Agreement context. First, we highlight the potential welfare distortions arising from trade liberalization, confirming theoretical predictions. The overall efficiency gains from trade liberalization may therefore be modest or even negative in oligopsonistic industries. Given that many modern industries operate under oligopsony conditions, these findings highlight the need for carefully designed trade policies. Second, our analysis illustrates the allocative and distributional effects of international trade within firms, driven by the increased exertion of labor market power. This emphasizes the importance of considering both efficiency and equity when evaluating the outcomes of trade liberalization.

References

- Abadie, A. (2005). Semiparametric Difference-in-Differences Estimators. *The Review of Economic Studies* 72(1), 1–19.
- Ackerberg, D. A., K. Caves, and G. Frazer (2015). Identification Properties of Recent Production Function Estimators. *Econometrica* 83(6), 2411–2451.
- Arkolakis, C., A. Costinot, D. Donaldson, and A. Rodríguez-Clare (2019). The elusive pro-competitive effects of trade. *The Review of Economic Studies* 86(1), 46–80.
- Arni, P., P. H. Egger, K. Erhardt, M. Gubler, and P. U. Sauré (2024). Heterogeneous Impacts of Trade Shocks on Workers.
- Atkeson, A. and A. Burstein (2008). Pricing-to-market, trade costs, and international relative prices. *American Economic Review* 98(5), 1998–2031.
- Autor, D., D. Dorn, L. F. Katz, C. Patterson, and J. Van Reenen (2020). The Fall of the Labor Share and the Rise of Superstar Firms. *The Quarterly Journal of Economics* 135(2), 645–709.
- Berger, D., K. Herkenhoff, and S. Mongey (2022). Labor market power. *American Economic Review* 112(4), 1147–1193.
- Bernard, A. B., T. C. Fort, V. Smeets, and F. Warzynski (2020). Heterogeneous globalization: Offshoring and reorganization. Technical report, National Bureau of Economic Research.
- Blundell, R. and S. Bond (1998). Initial Conditions and Moment Restrictions in Dynamic Panel Data Models. *Journal of Econometrics* 87(1), 115–143.
- Blundell, R. and S. Bond (2000). GMM Estimation with Persistent Panel Data: An Application to Production Functions. *Econometric Reviews* 19(3), 321–340.
- Bond, S., A. Hashemi, G. Kaplan, and P. Zoch (2021). Some Unpleasant Markup Arithmetic: Production Function Elasticities and their Estimation from Production Data. *Journal of Monetary Economics* 121, 1–14.
- Burstein, A., J. Cravino, and J. Vogel (2013). Importing Skill-Biased Technology. *American Economic Journal: Macroeconomics* 5(2), 32–71.
- Burstein, A. and J. Vogel (2017). International trade, technology, and the skill premium. *Journal of Political Economy* 125(5), 1356–1412.
- Caetano, C. and B. Callaway (2024). Difference-in-Differences when Parallel Trends Holds conditional on Covariates. Technical report, arXiv preprint arXiv:2406.15288.
- Callaway, B. and P. H. Sant’Anna (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics* 225(2), 200–230.
- Chou, C.-A. (2009). A Two-Edged Sword: The Economic Cooperation Framework Agreement between the Republic of China and the People’s Republic of China. *Int’l L. & Mgmt. Rev.* 6, 1.

- De Loecker, J., J. Eeckhout, and G. Unger (2020). The Rise of Market Power and the Macroeconomic Implications. *The Quarterly Journal of Economics* 135(2), 561–644.
- De Loecker, J., P. K. Goldberg, A. K. Khandelwal, and N. Pavcnik (2016). Prices, markups, and trade reform. *Econometrica* 84(2), 445–510.
- Deb, S., J. Eeckhout, A. Patel, and L. Warren (2024). Walras–Bowley Lecture: Market Power and Wage Inequality. *Econometrica* 92(3), 603–636.
- Dobbelaere, S. and J. Mairesse (2013). Panel Data estimates of the Production Function and Product and Labor Market Imperfections. *Journal of Applied Econometrics* 28(1), 1–46.
- Doraszelski, U. and J. Jaumandreu (2019). Using Cost Minimization to Estimate Markups. Technical report, CEPR Discussion Paper No. DP14114.
- Edmond, C., V. Midrigan, and D. Y. Xu (2015). Competition, markups, and the gains from international trade. *American Economic Review* 105(10), 3183–3221.
- Fan, J. and L. Li (2025). Skill-Biased Imports, Skill Acquisition, and Migration. *Journal of International Economics*, 104128.
- Felix, M. (2021). Trade, labor market concentration, and wages. *Job Market Paper* 64.
- Foster, L. S., J. C. Haltiwanger, and C. Tuttle (2022). Rising Markups or Changing Technology? Technical report, National Bureau of Economic Research.
- Grossman, G. M., E. Helpman, and P. Kircher (2017). Matching, sorting, and the distributional effects of international trade. *Journal of political economy* 125(1), 224–264.
- Gutiérrez, A. (2022). Labor market power and the pro-competitive gains from trade. *Unpublished Working Paper*.
- Helpman, E., O. Itskhoki, M.-A. Muendler, and S. J. Redding (2016). Trade and inequality: From theory to estimation. *The Review of Economic Studies* 84(1), 357–405.
- Helpman, E., O. Itskhoki, and S. Redding (2010). Inequality and unemployment in a global economy. *Econometrica* 78(4), 1239–1283.
- Holmes, T. J., W.-T. Hsu, and S. Lee (2014). Allocative efficiency, mark-ups, and the welfare gains from trade. *Journal of International Economics* 94(2), 195–206.
- Hong, T.-L. and C.-H. Yang (2011). The Economic Cooperation Framework Agreement between China and Taiwan: Understanding its Economics and Politics. *Asian Economic Papers* 10(3), 79–96.
- Korinek, A. (2023). Generative AI for Economic Research: Use Cases and Implications for Economists. *Journal of Economic Literature* 61(4), 1281–1317.
- Kusaka, S., T. Okazaki, K. Onishi, and N. Wakamori (2022). The Decline of the Labor Share and New Technology Diffusion. Technical report, mimeo, Federal Reserve Board.

- Lamadon, T., M. Mogstad, and B. Setzler (2022). Imperfect Competition, Compensating Differentials, and Rent Sharing in the US Labor Market. *American Economic Review* 112(1), 169–212.
- Levinsohn, J. and A. Petrin (2003). Estimating Production Functions Using Inputs to Control for Unobservables. *The Review of Economic Studies* 70(2), 317–341.
- Loecker, J. D. and F. Warzynski (2012). Markups and firm-level export status. *American economic review* 102(6), 2437–2471.
- Lu, Y., Y. Sugita, L. Zhu, et al. (2019). Wage and markdowns and FDI liberalization. Technical report, Hitotsubashi Institute for Advanced Study, Hitotsubashi University.
- MacKenzie, G. (2021). Trade and market power in product and labor markets. Technical report, Bank of Canada Staff Working Paper.
- Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *econometrica* 71(6), 1695–1725.
- Melitz, M. J. and G. I. Ottaviano (2008). Market size, trade, and productivity. *The review of economic studies* 75(1), 295–316.
- Miller, N., M. Osborne, G. Sheu, and G. Sileo (2023). Technology and Market Power: The United States Cement Industry, 1974-2019. *Georgetown McDonough School of Business Research Paper* (4041168).
- Olley, G. S. and A. Pakes (1996). The Dynamics of Productivity in the Telecommunications Equipment Industry. *Econometrica* 64(6), 1263–1297.
- Pham, H. (2023). Trade reform, oligopsony, and labor market distortion: Theory and evidence. *Journal of International Economics* 144, 103787.
- Rubens, M. (2023). Market Structure, Oligopsony power, and Productivity. *American Economic Review* 113(9), 2382–2410.
- Rubens, M., Y. Wu, and M. Xu (2024). Exploiting or Augmenting Labor? Technical report, Working Paper.
- Ruhl, K. J. and J. L. Willis (2017). New Exporter Dynamics. *International Economic Review* 58(3), 703–726.
- Sampson, T. (2014). Selection into trade and wage inequality. *American Economic Journal: Microeconomics* 6(3), 157–202.
- Sant’Anna, P. H. and J. Zhao (2020). Doubly Robust Difference-in-Differences Estimators. *Journal of Econometrics* 219(1), 101–122.
- Treuren, L. (2023). Wage Markups and Buyer Power in Intermediate Input Markets. Technical report.

Trottner, F. (2019). Who gains from scale? Technical report, Working Paper.

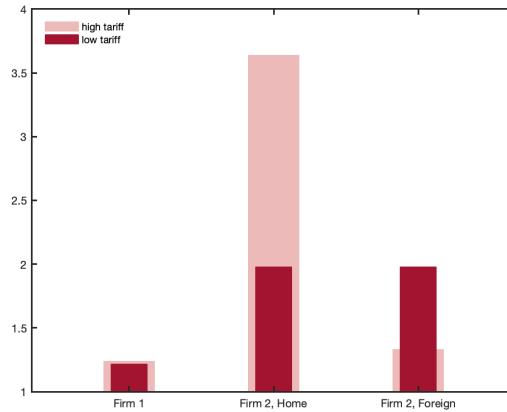
Xie, E., M. Xu, and M. Yu (2024). Trade liberalization, labor market power, and misallocation across firms: Evidence from China's WTO accession. *Journal of Development Economics* 171, 103353.

Yeh, C., C. Macaluso, and B. Hershbein (2022). Monopsony in the US Labor Market. *American Economic Review* 112(7), 2099–2138.

A Conceptual Framework

Product markups. In Figure A1, we present the responses of firm-level markups to trade liberalization. In the low-tariff equilibrium, increased foreign competition leads to a significant decline in firm 2's domestic markup, generating pro-competitive effects of trade, consistent with Edmond et al. (2015). However, the lower tariff also makes firm 2 more competitive in the foreign market, allowing it to raise its foreign markup. This finding aligns with the empirical evidence in Loecker and Warzynski (2012), who report that firm-level markups tend to increase upon export entry.

Figure A1: A Qualitative Example: Markup Responses



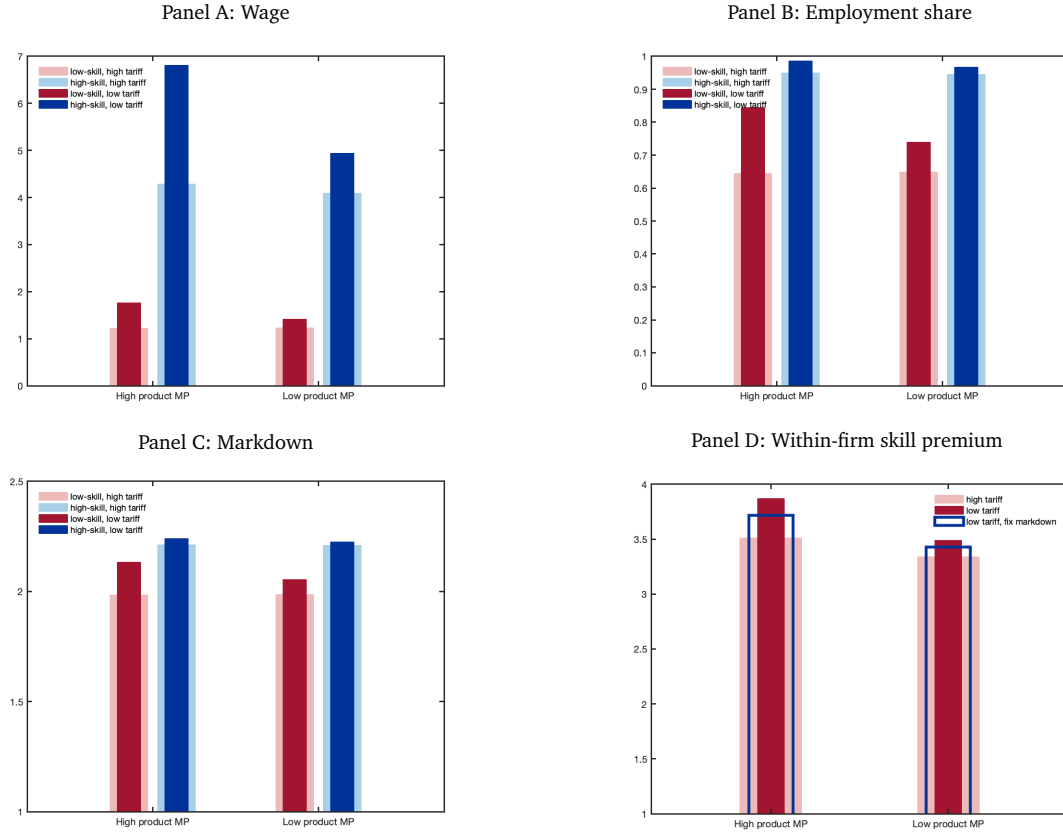
Note: The figure displays the effect of trade liberalization on product markup. The baseline equilibrium involve high tariff. The trade liberalization equilibrium refers to the one with low tariff. Firm 1 serves only the domestic market, whereas Firm 2 serves both the Home and Foreign markets.

Interaction between product and labor market power. The degree of product market power influences the magnitude of resource reallocation following trade liberalization, which is ultimately linked to the response of wage markdowns. As a robustness check, here we vary the degree of product market power. Specifically, we adjust the relative magnitudes of the within-

and across-market elasticities of substitution in the product markets, γ and θ . As shown in Equation (4), holding γ constant, an increase in θ causes the product markup to rise more slowly with respect to market share. This reduces the sensitivity of markups to market share, thereby weakening product market power. To assess the implications, we increase θ and adjust firm productivity so that, in the high-tariff equilibrium, firm 2 still employs 95% of high-skilled labor and 65% of low-skilled labor within a market.¹⁴ All other parameters remain unchanged. Figure A2 presents the results, focusing on firm 2. As markups become less responsive to market share, labor reallocation following trade liberalization becomes more limited. This in turn dampens the increase in wage markdowns and within-firm wage inequality. Nonetheless, the qualitative patterns persist: low-skilled labor experiences stronger reallocation and a larger increase in wage markdowns, which contributes to rising wage inequality within firm 2.

¹⁴We increase θ from 2 to 2.5. When $\gamma = \theta$, the markup becomes constant, as in the monopolistic competition framework of Melitz (2003).

Figure A2: A Qualitative Example: High vs Low Product Market Power



Note: The figure displays the effect of trade liberalization on wage, employment share, wage markdown and within-firm skill premium for the high-productivity firm 2. The baseline equilibrium involves high tariff. The trade liberalization equilibrium refers to the one with low tariff. High product market power (MP) refers to the case where $\gamma = 6$ and $\theta = 1.2$, whereas low product market power corresponds to $\gamma = 6$ and $\theta = 2.5$.

B Deriving Firms' Wage Markdowns

Production

In period t , Taiwanese machinery manufacturer j produces Y_{jt} units of product using four production factors: low-skilled labor L_{jt} , high-skilled labor H_{jt} , raw material M_{jt} , and fixed physical capital K_{jt} . Firms in a given sector are assumed to face a common technological constraint, characterized by a continuously differentiable production function in all four production factors:

$$Y_{jt} = F_{jt}(L_{jt}, H_{jt}, M_{jt}, K_{jt}),$$

which can exhibit flexible output elasticities across firms and times. We denote the output elasticity of low-skilled labor, high-skilled labor, and material input by θ_{jt}^L , θ_{jt}^H , and θ_{jt}^M :

$$\theta_{jt}^L = \frac{\partial F(\cdot)}{\partial L_{jt}} \frac{L_{jt}}{Y_{jt}}; \quad \theta_{jt}^H = \frac{\partial F(\cdot)}{\partial H_{jt}} \frac{H_{jt}}{Y_{jt}}; \quad \theta_{jt}^M = \frac{\partial F(\cdot)}{\partial M_{jt}} \frac{M_{jt}}{Y_{jt}}. \quad (\text{B.1})$$

Input Markets and Wage Markdowns

We assume low-skilled labor L_{jt} , high-skilled labor H_{jt} , and raw material M_{jt} are static inputs. Let W_{jt}^L , W_{jt}^H , and W_{jt}^M be wage rates for low-skilled, wage rates for high-skilled, and material prices, respectively. We assume that manufacturer j faces a finitely elastic labor supply curve, whose elasticity is denoted by ψ_{jt}^X , $X \in \{L, H\}$:

$$\psi_{jt}^X = \frac{\partial X_{jt}}{\partial W_{jt}^X} \frac{W_{jt}^X}{X_{jt}} > 0. \quad (\text{B.2})$$

In contrast, all the manufacturers take the price of raw material as given, implying

$$\psi_{jt}^M = \frac{\partial M_{jt}}{\partial W_{jt}^M} \frac{W_{jt}^M}{M_{jt}} = 0. \quad (\text{B.3})$$

Under the standard profit maximization assumption, manufacturer j 's wage markdowns δ_{jt}^L and δ_{jt}^H depend on the inverse of the perceived elasticity of labor:

$$\frac{MPRL_{jt}^X}{W_{jt}^X} = \delta_{jt}^X = 1 + \frac{1}{\psi_{jt}^X}, \quad (\text{B.4})$$

where $MPRL_{jt}^X$ is marginal revenue product of labor type $X \in \{L, H\}$ for manufacturer j .

Behavioral Assumption

Manufacturer j chooses its optimal low-skilled labor, high-skilled labor, and raw material by engaging in short-run cost minimization.

$$\min_{L_{jt}, H_{jt}, M_{jt}} W_{jt}^L L_{jt} + W_{jt}^H H_{jt} + W_{jt}^M M_{jt} \quad \text{s.t.} \quad F_{jt}(L_{jt}, H_{jt}, M_{jt}, K_{jt}) \geq Y_{jt}. \quad (\text{B.5})$$

With some rearrangements of the terms, the first-order conditions for the cost minimization with respect to labor and raw material can be given by:

$$W_{jt}^L + \frac{\partial W_{jt}^L}{\partial L_{jt}} L_{jt} = \frac{\lambda_{jt}}{P_{jt}} \theta_{jt}^L \frac{R_{jt}}{L_{jt}}, \quad (\text{B.6})$$

$$W_{jt}^H + \frac{\partial W_{jt}^H}{\partial H_{jt}} H_{jt} = \frac{\lambda_{jt}}{P_{jt}} \theta_{jt}^H \frac{R_{jt}}{H_{jt}}, \quad (\text{B.7})$$

$$W_{jt}^M = \frac{\lambda_{jt}}{P_{jt}} \theta_{jt}^M \frac{R_{jt}}{M_{jt}}, \quad (\text{B.8})$$

where λ_{jt} is the Lagrangian multiplier associated with cost minimization problem (B.5), and R_{jt} and P_{jt} are firm-level revenues and output prices. Combining equations (B.6), (B.7), and (B.8) and rearranging the terms, we obtain

$$1 + \underbrace{\frac{\partial W_{jt}^L}{\partial L_{jt}} \frac{L_{jt}}{W_{jt}^L}}_{=\frac{1}{\psi_{jt}^L}} = \frac{\theta_{jt}^L}{\theta_{jt}^M} \frac{W_{jt}^M M_{jt}}{W_{jt}^L L_{jt}} \quad (\text{B.9})$$

$$1 + \underbrace{\frac{\partial W_{jt}^H}{\partial H_{jt}} \frac{H_{jt}}{W_{jt}^H}}_{=\frac{1}{\psi_{jt}^H}} = \frac{\theta_{jt}^H}{\theta_{jt}^M} \frac{W_{jt}^M M_{jt}}{W_{jt}^H H_{jt}} \quad (\text{B.10})$$

Due to the duality between profit maximization and cost minimization problems, the left-hand sides of equations (B.9) and (B.10) represent wage markdowns over low-skilled and high-skilled workers, respectively as (B.4). We thus identify firm-level wage markdowns using equations (B.9) and (B.10), which requires estimation of the production function to identify θ_{jt}^L , θ_{jt}^H , and θ_{jt}^M .

C Production Function Estimation

C.I System Generalized Method of Moments

Recall a translog specification for production technology (10):

$$\begin{aligned}
 y_{jt} = & \beta_l l_{jt} + \beta_h h_{jt} + \beta_m m_{jt} + \beta_k k_{jt} + \sum_{\{n,o\} \in \{l,h,m,k\}} \sum \beta_{no} n_{jt} o_{jt} \\
 & + \left(\beta_l^{SBT} l_{jt} + \beta_h^{SBT} h_{jt} + \beta_m^{SBT} m_{jt} + \beta_k^{SBT} k_{jt} + \sum_{\{n,o\} \in \{l,h,m,k\}} \sum \beta_{no}^{SBT} n_{jt} o_{jt} \right) s_{jt} \\
 & + \omega_{jt} + \varepsilon_{jt}, \\
 \equiv & \mathcal{X}'_{jt} \beta + \omega_{jt} + \varepsilon_{jt}
 \end{aligned} \tag{C.11}$$

where all $\mathcal{X}_{jt}$ is a collection of all observed variables.

We assume logged TFP ω_{jt} consists of fixed TFP ω_j^* and time-varying productivity term $\tilde{\omega}_{jt}$, and $\tilde{\omega}_{jt}$ evolves according to the AR(1) process:

$$\begin{aligned}
 \omega_{jt} &= \omega_j^* + \tilde{\omega}_{jt}, \\
 \tilde{\omega}_{jt} &= \rho \tilde{\omega}_{j,t-1} + \xi_{jt},
 \end{aligned} \tag{C.12}$$

where ξ_{jt} is idiosyncratic productivity shocks observed to firm j but unobserved to the econometrician. ε_{jt} is the output measurement errors.

The ordinary least squares (OLS) estimator of equation (C.11) suffers from simultaneity bias due to the endogenous relationship between static production factors (l_{jt}, m_{jt}) and productivity ω_{jt} . Widely adopted approaches to address this issue, often referred to as "control

function" methods, are introduced and further developed by [Olley and Pakes \(1996\)](#), [Levinsohn and Petrin \(2003\)](#), and [Akerberg et al. \(2015\)](#). These methods rely on either implicit or explicit assumptions of market conduct and adopt the scalar unobservable assumption (i.e., productivity is the sole unobservable factor) to simplify estimation ([Doraszelski and Jaumandreu \(2019\)](#)). However, such methods are incompatible with our oligopolistic and oligopsonistic framework, which incorporates multiple unobservables: productivity, price markups, and wage markdowns.

To address these challenges, we build upon the “dynamic panel” approach proposed by [Blundell and Bond \(2000\)](#), which requires only the standard timeline assumption of input choices and an AR(1) process for logged productivity. In our application, we assume that Taiwanese manufacturer j chooses its capital k_{jt} before observing productivity shock ξ_{jt} , while choosing static production factors (l_{jt}, m_{jt}) after observing ξ_{jt} . This information assumption is widely adopted in previous empirical studies on firm productivity. In addition, we consider a stationary economy in which the underlying primitives of the economy, such as demand, production technology, and investment cost structure, are time-invariant.

To exploit the imposed assumptions, we first quasi-difference production function (C.11) by using the Markovian structure of productivity in equation (C.12):

$$\underbrace{\xi_{jt} + (1 - \rho)\omega_j^* + \varepsilon_{jt} - \rho\varepsilon_{jt-1}}_{\zeta_{jt}(\beta, \rho)} = y_{jt} - \rho y_{jt-1} - (\mathcal{X}_{jt} - \rho \mathcal{X}_{jt-1})' \beta. \quad (\text{C.13})$$

We then difference out ω_j^* by within-firm first differencing equation (C.13):

$$\underbrace{\xi_{jt} - \xi_{jt-1} + \varepsilon_{jt} - (1 - \rho)\varepsilon_{jt-1} + \rho\varepsilon_{jt-2}}_{\Delta\zeta_{jt}(\beta, \rho)} = \Delta y_{jt} - \rho\Delta y_{jt-1} - (\Delta\mathcal{X}_{jt} - \rho\Delta\mathcal{X}_{jt-1})'\beta. \quad (\text{C.14})$$

We perform the method of generalized moments (GMM) to estimate (β, ρ) by constructing the moment condition based on the information and stationary assumptions. First, we break down the production function variables into static and fixed parts:

$$\begin{aligned} \mathcal{X}_{jt}^v &= \left((l_{jt}, l_{jt}^2, h_{jt}, h_{jt}^2, m_{jt}, m_{jt}^2, l_{jt}m_{jt}, l_{jt}h_{jt}, h_{jt}m_{jt}), (l_{jt}, l_{jt}^2, h_{jt}, h_{jt}^2, m_{jt}, m_{jt}^2, l_{jt}m_{jt}, l_{jt}h_{jt}, h_{jt}m_{jt}) \times s_{jt} \right), \\ \mathcal{X}_{jt}^f &= \left((k_{jt}, k_{jt}^2, l_{jt-1}k_{jt}, h_{jt-1}k_{jt}, m_{jt-1}k_{jt}), (k_{jt}, k_{jt}^2, l_{jt-1}k_{jt}, h_{jt-1}k_{jt}, m_{jt-1}k_{jt}) \times s_{jt} \right). \end{aligned}$$

The information assumption implies that the current productivity shock is uncorrelated with all the past labor and material flows and all the current and past capital stocks. Thus, the moment condition is given by

$$\mathbb{E} \left[\Delta\zeta_{jt}(\beta, \rho) \mid \{\mathcal{X}_{jt-b-1}^v, \mathcal{X}_{jt-b}^f\}_{b \geq 2} \right] = 0, \quad \forall t = 4, 5, \dots, T. \quad (\text{C.15})$$

Due to the stationary assumption, the parameters of conditional factor demand functions for labor and material and the investment policy function are time-invariant. Hence, within-firm first differences in input factors are not correlated to fixed TFP ω_j^* . The moment condition is

$$\mathbb{E} \left[\zeta_{jt}(\beta, \rho) \mid \{\Delta\mathcal{X}_{jt-b-1}^v, \Delta\mathcal{X}_{jt-b}^f\}_{b=1} \right] = 0, \quad \forall t = 3, 4, \dots, T. \quad (\text{C.16})$$

We search for (ρ, β) that best rationalizes the system of moments conditions (C.15) and (C.16) via the two-step GMM technique.

C.II Estimation Results of Production Function

Table C1 presents the structural estimates of the baseline production function parameters.

Table C2 presents the structural estimates of the Skill-Biased Technical Changes (SBTC) production function parameters.

Table C1: Production Function Estimates:
Baseline Parameters

	OLS		GMM	
	Estimate	S.E.	Estimate	S.E.
ρ			0.6125	0.0178
β_l	0.1876	0.0036	0.1925	0.0093
β_{ll}	0.0559	0.0031	0.0490	0.0058
β_h	0.0728	0.0029	0.0544	0.0061
β_{hh}	0.0147	0.0011	0.0124	0.0020
β_m	0.6712	0.0280	0.6522	0.0072
β_{mm}	0.0744	0.0023	0.0644	0.0043
β_k	0.0511	0.0010	0.0591	0.0022
β_{kk}	0.0112	0.0033	0.0061	0.0067
β_{lh}	0.0011	0.0049	-0.0039	0.0067
β_{lm}	-0.1052	0.0049	-0.0877	0.0093
β_{lk}	0.0013	0.0027	0.0117	0.0065
β_{hm}	-0.0171	0.0027	-0.0241	0.0055
β_{hk}	0.0019	0.0017	0.0032	0.0037
β_{mk}	-0.0281	0.0021	-0.0270	0.0042
N	38,610		29,697	

Notes: This table reports the estimates of the production function parameters. All the specifications control for the time fixed effects. Standard errors in parenthesis are clustered at the firm-level.

Table C2: Production Function Estimates:
SBTC Parameters

	GMM	
	Estimate	S.E.
β_l^{SBT}	0.0003	0.0016
β_{ll}^{SBT}	0.0005	0.0006
β_h^{SBT}	0.0098	0.0012
β_{hh}^{SBT}	0.0006	0.0004
β_m^{SBT}	-0.0085	0.0013
β_{mm}^{SBT}	0.0031	0.0005
β_k^{SBT}	0.0007	0.0007
β_{kk}^{SBT}	0.0002	0.0002
β_{lh}^{SBT}	-0.0001	0.0009
β_{lm}^{SBT}	-0.0010	0.0011
β_{lk}^{SBT}	0.0001	0.0004
β_{hm}^{SBT}	-0.0034	0.0007
β_{hk}^{SBT}	0.0022	0.0003
β_{mk}^{SBT}	-0.0021	0.0004
N	29,697	

Notes: This table reports the estimates of the production function parameters. All the specifications control for the time fixed effects. Standard errors in parenthesis are clustered at the firm-level.